

Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System

CO5-ASSESSMENT 1

Course:

Software Engineering / Code Review and Version Control Evaluation

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RESEARCH PAPER REVIEW / LITERATURE SURVEY

Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System

Literature survey covering AI, IoT, smart-room technologies, conversational agents, service robots, recommendation systems, and intelligent hotel operations

Abstract

This literature survey reviews research relevant to an Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System. The review focuses on five interconnected capabilities: intelligent hotel operations, smart-room and IoT integration, conversational/voice assistance, AI and robotic service delivery, and personalized recommendations and feedback analytics. Recent peer-reviewed work shows that AI can improve personalization, service responsiveness, operational efficiency, and decision support, but the literature remains fragmented across individual use cases. The strongest research gap is the lack of an end-to-end, interoperable, privacy-aware platform that combines real-time operational data with guest-facing AI and evaluates the resulting system using both technical metrics and longitudinal guest/service outcomes. The survey therefore motivates a unified architecture combining IoT sensing, hotel-management-system integration, machine learning, NLP/LLM-based assistance, recommendation, predictive analytics, and human-in-the-loop service recovery.

1. Introduction

Hotels are increasingly becoming cyber-physical service environments in which reservations, rooms, energy systems, access control, customer communication, housekeeping, service requests, and feedback generate interconnected data. Research on smart hospitality shows that AI, IoT, big-data analytics, voice interfaces, and robots can transform both the back-office and the guest-facing service encounter. However, individual studies usually investigate one component—such as technology acceptance, a service robot, a voice assistant, or a recommender—rather than an integrated hotel operations platform.

This survey consequently treats the assigned problem as a systems-integration problem as well as an AI problem. The target system should not merely automate tasks; it should use heterogeneous hotel data to anticipate needs, personalize interactions, optimize resources, and escalate complex cases to human staff.

2. Review Method and Paper Selection

The literature was selected from peer-reviewed journals and scholarly conference proceedings indexed or published by major academic publishers such as Elsevier, Springer Nature, Emerald, and related scholarly repositories. Priority was given to papers directly addressing smart hotels, AI/robotic service, guest experience, personalization/recommendation, IoT-enabled hotel operations, and empirical evaluation. Papers were also selected to provide methodological diversity: surveys and structural models, action research, systematic reviews, conceptual architectures, machine-learning experiments, and recommendation models.

Search themes: smart hotel, artificial intelligence in hospitality, hotel IoT, guest experience, service robots, voice assistants, hotel recommendation, hotel operations, and AI-based guest satisfaction analysis.

Scope note: the review emphasizes work available by 22 August 2026; therefore, later-dated publications are not treated as available evidence.

3. Individual Paper Reviews

3.1 Citak, Owoc & Weichbroth (2021)

Research problem: AI applications in hospitality and hotel adoption.

Methodology/dataset: State-of-the-art survey plus questionnaire; n=178.

Technologies/tools: AI applications, chatbots/messaging, ML business intelligence, VR/AR; survey analysis.

Major findings: AI is already used across customer-facing services and analytics, but adoption and guest-experience evidence remain incomplete.

Limitations: Small perception sample; broad survey rather than a deployed hotel system; limited quantitative operational evaluation.

Implication for the proposed system: Need integrated, measurable hotel AI systems and stronger evaluation of benefits, risks, adoption, and guest outcomes.

3.2 Buhalis & Moldavska (2021)

Research problem: In-room voice-based AI assistants for hotel services.

Methodology/dataset: Conceptual/industry-focused analysis of hotel voice assistants.

Technologies/tools: Voice AI, speech interaction, IoT/smart-room integration.

Major findings: Voice assistants create a new guest interaction touchpoint and can automate routine room/service requests.

Limitations: Deployment was still limited; technical imperfections can create dissatisfaction; privacy, trust, and interoperability issues remain.

Implication for the proposed system: Need production-scale, multilingual, context-aware assistants integrated with hotel workflows and evaluated with real service data.

3.3 Mingotto, Montaguti & Tamma (2021)

Research problem: AI/robotics in frontline hotel operations.

Methodology/dataset: Action research case study using Pepper robot at an Italian hotel.

Technologies/tools: Humanoid robot, supervised ML conversational AI, observation, questionnaires.

Major findings: Robot performance improved over time: exact answers increased from about 51.57% to over 75.33%; smooth interaction and satisfaction also improved.

Limitations: Single hotel/case; narrow information-service task; AI remained weak for complex requests and required employee support.

Implication for the proposed system: Need robust human-in-the-loop architectures, broader tasks, multi-hotel validation, and continuous monitoring of accuracy and service recovery.

3.4 Yang, Song, Cheung & Guan (2021)

Research problem: Guest acceptance of smart-hotel technology.

Methodology/dataset: Extended Technology Acceptance Model; online survey; 648 valid responses.

Technologies/tools: TAM, technology readiness, perceived usefulness/ease of use, structural analysis.

Major findings: Technology readiness affected intention to visit smart hotels; perceived usefulness/ease of use related to technology amenities.

Limitations: Self-reported intention rather than actual long-term usage; no integrated operational evaluation; contextual and demographic generalizability limits.

Implication for the proposed system: Need systems that connect acceptance research with actual usage telemetry, service quality, personalization, and operational performance.

3.5 Brylska, Cobanoglu & Dogan (2022)

Research problem: Guest attitudes toward hotel service robots.

Methodology/dataset: Self-administered survey; 939 usable hotel-guest responses; factor analysis.

Technologies/tools: Scale development/factor analysis; robot attitudes, advantages, disadvantages, fear and pandemic-related factors.

Major findings: Five dimensions emerged; guests recognized efficiency/experience benefits while also expressing concerns and fear; users and non-users differed.

Limitations: Cross-sectional self-report; attitudes do not necessarily equal real-world performance; robot type and service context may vary.

Implication for the proposed system: Need contextual, longitudinal evaluation of robot + human service combinations and measurable operational/experience outcomes.

3.6 Aggarwal & Mittal (2024)

Research problem: Autonomous end-to-end smart hotel architecture.

Methodology/dataset: Conceptual modelling based on literature review.

Technologies/tools: DASH: AI + ML + IoT + robots + blockchain; pay-per-use model.

Major findings: Proposes a decentralized autonomous smart-hotel concept emphasizing trust, transparency, utility tracking, and operational efficiency.

Limitations: Conceptual rather than experimentally validated; integration, latency, security, cost, and organizational feasibility are not fully quantified.

Implication for the proposed system: Need prototype implementation and empirical benchmarking of integrated architecture, interoperability, security, and ROI.

3.7 Pu, Xu, Zhang, Zeng & Gan (2024)

Research problem: Personalized hotel recommendation from online reviews.

Methodology/dataset: Two-stage recommendation model; TripAdvisor case study.

Technologies/tools: LDA topic extraction, sentiment analysis, risk-attitude filtering, probabilistic linguistic term-set cosine similarity.

Major findings: Outperformed comparison recommendation models and incorporated personalized preferences and customer risk attitudes.

Limitations: Depends on review quality/history; cold-start and real-time context are difficult; recommendation is not tightly integrated with in-stay operations.

Implication for the proposed system: Need real-time, multimodal, context-aware recommendations using current stay data, room/IoT signals, and service interactions.

3.8 Diwan (2025)

Research problem: Centralized smart hotel operations and guest satisfaction.

Methodology/dataset: Integrated system design with ML and multi-criteria decision analysis.

Technologies/tools: IoT, face recognition (Viola-Jones/Haar Cascade), MySQL, CSS/Bootstrap/Java Swing, Random Forest, DNN, fuzzy-AHP.

Major findings: The centralized system combines reservations, checkout, settlements, access/check-in and guest analytics; Random Forest showed lower MSE/RMSE than DNN in the reported satisfaction modelling.

Limitations: Face-recognition and centralized architecture raise privacy/security and scalability questions; evaluation appears limited to the proposed environment.

Implication for the proposed system: Need privacy-preserving identity, interoperable PMS/IoT integration, scalable cloud/edge architecture, and external validation across hotels.

3.9 Du, Li, So & King (2025)

Research problem: Consumer receptivity to unmanned smart hotels.

Methodology/dataset: Two sequential studies; 250 valid survey responses in Study 2.

Technologies/tools: TAM/TPB-based model; perceived security, physical risk, privacy, website design, hotel reputation.

Major findings: Attitude, subjective norms and perceived behavioral control predicted intention; perceived security was shaped by physical risk, privacy concerns, design and reputation.

Limitations: Sample heavily concentrated among young students; intention-based rather than longitudinal real usage; security was studied mainly from the consumer perspective.

Implication for the proposed system: Need privacy-by-design, transparent AI, secure identity/access, and field studies linking security controls to real guest experience.

3.10 Cheng, Liu & Tussyadiah (2026)

Research problem: Customer experience in smart hotels.

Methodology/dataset: Systematic review following PRISMA; 35 empirical studies published 2014–2024.

Technologies/tools: Systematic literature review and conceptual framework.

Major findings: Identifies antecedents, mediators, moderators and outcomes of smart-hotel customer experience; highlights the complexity of technology-enabled experience.

Limitations: Review is dependent on the scope/quality of existing empirical work; relatively few studies connect customer experience to end-to-end operational data.

Implication for the proposed system: Need integrated empirical studies that jointly measure AI performance, operational KPIs, guest emotions/satisfaction, and human-service outcomes.

3.11 Sim & Kim (2026)

Research problem: Which AI-robot service attributes matter across hotel contexts.

Methodology/dataset: Cross-cultural Kano-model study with HRI theory; U.S. and Korean guests; luxury vs midscale; AI-only vs AI-human collaboration.

Technologies/tools: Kano model, human-robot interaction theory, cross-cultural comparison.

Major findings: Service context was more influential than hotel type; technology readiness moderated evaluations, especially in luxury settings.

Limitations: Focuses on evaluation of service attributes rather than full operational integration; cultural/context effects require broader validation.

Implication for the proposed system: Need adaptive orchestration that chooses AI-only, AI-human, or human service according to task complexity, context, culture, and guest readiness.

3.12 De Mena-Fernández de Larrea, Aramendia-Muneta & Martínez-Ruiz (2026)

Research problem: Customer experience with service robots.

Methodology/dataset: Scenario-based questionnaire; data collected in Spain during Sep–Oct 2024; 5-point Likert scales.

Technologies/tools: Service-robot quality, brand reputation, attitudes, customer experience; USUS-model framing.

Major findings: Examines how robot quality can affect experience through brand reputation and attitudes; scenario included check-in/out, facial recognition, recommendations, delivery, multilingual Q&A.

Limitations: Hypothetical scenario rather than direct longitudinal deployment; one national context; self-report measures.

Implication for the proposed system: Need live hotel deployments combining service quality, task success, guest outcomes, and operational telemetry.

4. Comparative Analysis

The studies can be grouped into four streams. First, acceptance/experience studies explain whether guests are willing to use smart technologies. Second, robot and voice-agent studies evaluate specific interaction channels. Third, recommendation and analytics studies demonstrate personalization from reviews or structured data. Fourth, architecture/system studies show how IoT, AI, robotics and enterprise systems can be combined. The streams are complementary but are rarely evaluated together.

Author /year	Research problem	Methodology	Dataset/tools	Key findings	Limitations	Research gap	Primary metric/evidence
Citak, Owoc & Weichbroth (2021)	AI applications in hospitality and hotel adoption	State-of-the-art survey plus questionnaire; n=178	AI applications, chatbots/messaging, ML business intelligence, VR/AR; survey analysis	AI is already used across customer-facing services and analytics, but adoption and guest-experience evidence remain incomplete.	Small perception sample; broad survey rather than a deployed hotel system; limited quantitative operational evaluation.	Need integrated, measurable hotel AI systems and stronger evaluation of benefits, risks, adoption, and guest outcomes.	Survey/factor analysis
Buhalis & Moldavska (2021)	In-room voice-based AI assistants for hotel services	Conceptual/industry-focused analysis of hotel voice assistants	Voice AI, speech interaction, IoT/smart-room integration	Voice assistants create a new guest interaction touchpoint and can automate routine room/service requests.	Deployment was still limited; technical imperfections can create dissatisfaction; privacy, trust, and interoperability issues remain.	Need production-scale, multilingual, context-aware assistants integrated with hotel workflows and evaluated with real service data.	Conceptual/system framework
Mingot	AI/robotic	Action	Humanoid	Robot	Single	Need robust	Empirical

to, Montaguti & Tamma (2021)	s in frontline hotel operations	research case study using Pepper robot at an Italian hotel	robot, supervised ML conversation al AI, observation, questionnaires	performance improved over time: exact answers increased from about 51.57% to over 75.33%; smooth interaction and satisfaction also improved.	hotel/case ; narrow information-service task; AI remained weak for complex requests and required employee support.	human-in-the-loop architectures, broader tasks, multi-hotel validation, and continuous monitoring of accuracy and service recovery.	case metrics
Yang, Song, Cheung & Guan (2021)	Guest acceptance of smart-hotel technology	Extended Technology Acceptance Model; online survey; 648 valid responses	TAM, technology readiness, perceived usefulness/ease of use, structural analysis	Technology readiness affected intention to visit smart hotels; perceived usefulness/ease of use related to technology amenities.	Self-reported intention rather than actual long-term usage; no integrated operational evaluation ; contextual and demographic generalizability limits.	Need systems that connect acceptance research with actual usage telemetry, service quality, personalization, and operational performance.	Survey/factor analysis
Brylska , Cobanoglu & Dogan (2022)	Guest attitudes toward hotel service robots	Self-administered survey; 939 usable hotel-guest responses; factor analysis	Scale development/factor analysis; robot attitudes, advantages, disadvantages, fear and pandemic-related factors	Five dimensions emerged; guests recognized efficiency/experience benefits while also expressing concerns and fear; users and non-users differed.	Cross-sectional self-report; attitudes do not necessarily equal real-world performance; robot type and service context may vary.	Need contextual, longitudinal evaluation of robot + human service combinations and measurable operational/experience outcomes.	Survey/factor analysis
Aggarwal	Autonomous	Conceptual	DASH: AI +	Proposes a	Conceptual	Need	Conceptual

al & Mittal (2024)	us end-to-end smart hotel architecture	modelling based on literature review	ML + IoT + robots + blockchain; pay-per-use model	decentralized autonomous smart-hotel concept emphasizing trust, transparency, utility tracking, and operational efficiency.	I rather than experimentally validated; integration, latency, security, cost, and organizational feasibility are not fully quantified.	prototype implementation and empirical benchmarking of integrated architecture, interoperability, security, and ROI.	/system framework
Pu, Xu, Zhang, Zeng & Gan (2024)	Personalized hotel recommendation from online reviews	Two-stage recommendation model; TripAdvisor case study	LDA topic extraction, sentiment analysis, risk-attitude filtering, probabilistic linguistic term-set cosine similarity	Outperformed comparison recommendation models and incorporated personalized preferences and customer risk attitudes.	Depends on review quality/history; cold-start and real-time context are difficult; recommendation is not tightly integrated with in-stay operations.	Need real-time, multimodal, context-aware recommendations using current stay data, room/IoT signals, and service interactions.	Model comparison / error metrics
Diwan (2025)	Centralized smart hotel operations and guest satisfaction	Integrated system design with ML and multi-criteria decision analysis	IoT, face recognition (Viola-Jones/Haar Cascade), MySQL, CSS/Bootstrap/Java Swing, Random Forest, DNN, fuzzy-AHP	The centralized system combines reservations, checkout, settlements, access/check-in and guest analytics; Random Forest showed lower MSE/RMSE than DNN in the reported satisfaction	Face-recognition and centralized architecture raise privacy/security and scalability questions; evaluation appears limited to the proposed environment.	Need privacy-preserving identity, interoperable PMS/IoT integration, scalable cloud/edge architecture, and external validation across hotels.	Conceptual /system framework

				modelling.			
Du, Li, So & King (2025)	Consumer receptivity to unmanne d smart hotels	Two sequential studies; 250 valid survey responses in Study 2	TAM/TPB-based model; perceived security, physical risk, privacy, website design, hotel reputation	Attitude, subjective norms and perceived behavioral control predicted intention; perceived security was shaped by physical risk, privacy concerns, design and reputation.	Sample heavily concentra ted among young students; intention-based rather than longitudin al real usage; security was studied mainly from the consumer perspectiv e.	Need privacy-by-design, transparent AI, secure identity/acces s, and field studies linking security controls to real guest experience.	Survey/fact or analysis
Cheng, Liu & Tussya diah (2026)	Customer experienc e in smart hotels	Systematic review following PRISMA; 35 empirical studies published 2014–2024	Systematic literature review and conceptual framework	Identifies antecedents, mediators, moderators and outcomes of smart-hotel customer experience; highlights the complexity of technology-enabled experience.	Review is dependen t on the scope/qua lity of existing empirical work; relatively few studies connect customer experienc e to end-to-end operation al data.	Need integrated empirical studies that jointly measure AI performance, operational KPIs, guest emotions/sati sfaction, and human-service outcomes.	Conceptual /system framework
Sim & Kim (2026)	Which AI-robot service attributes matter across hotel contexts	Cross-cultural Kano-model study with HRI theory; U.S. and Korean guests; luxury vs midscale; AI-only vs AI-	Kano model, human-robot interaction theory, cross-cultural comparison	Service context was more influential than hotel type; technology readiness moderated evaluations, especially in	Focuses on evaluation of service attributes rather than full operation al integratio n;	Need adaptive orchestration that chooses AI-only, AI-human, or human service according to task complexity, context, culture, and	Conceptual /system framework

		human collaboration		luxury settings.	cultural/context effects require broader validation.	guest readiness.	
De Mena-Fernández de Larrea, Aramendia-Muneta & Martínez-Ruiz (2026)	Customer experience with service robots	Scenario-based questionnaire; data collected in Spain during Sep–Oct 2024; 5-point Likert scales	Service-robot quality, brand reputation, attitudes, customer experience; USUS-model framing	Examines how robot quality can affect experience through brand reputation and attitudes; scenario included check-in/out, facial recognition, recommendations, delivery, multilingual Q&A.	Hypothetical scenario rather than direct longitudinal deployment; one national context; self-report measures.	Need live hotel deployments combining service quality, task success, guest outcomes, and operational telemetry.	Conceptual/system framework

Important comparison: direct numerical accuracy cannot be fairly compared across all papers because the tasks differ. For example, robot answer accuracy, recommendation quality, satisfaction-model error, technology-acceptance coefficients, and guest-experience ratings are not interchangeable. A future integrated system should therefore report a common KPI framework covering both technical and service outcomes.

5. Identified Limitations and Challenges in Existing Research

- Fragmented solutions: most studies optimize a single function rather than the complete hotel service lifecycle.
- Limited real-world longitudinal evidence: many studies use surveys, hypothetical scenarios, conceptual models, or one-hotel case studies.
- Interoperability: AI components, IoT devices, PMS/booking systems, access-control systems, housekeeping systems, and CRM platforms often lack a common integration layer.
- Context awareness: current systems struggle to combine guest profile, current request, room state, time, occupancy, service workload, and hotel policies into one decision.
- Human-AI collaboration: robots and conversational systems are useful for routine tasks but still require escalation and recovery mechanisms for ambiguous or high-value requests.
- Privacy and security: facial recognition, voice data, behavioral profiles, room sensors, and personalized recommendations create substantial privacy and cybersecurity requirements.
- Evaluation inconsistency: studies use different measures, making it difficult to compare accuracy, efficiency, satisfaction, cost, and scalability.

- Cold-start and data sparsity: recommendation systems may lack sufficient history for new guests and may not adapt quickly to changing preferences.
- Multilingual and accessibility challenges: guest-facing AI must handle accents, languages, disabilities, and different levels of technology readiness.
- Explainability and trust: guests and staff need understandable reasons for recommendations, automated decisions, alerts, and service escalations.

6. Research Gap

The central research gap is the absence of a validated, end-to-end intelligent hotel platform that unifies operational intelligence and guest-experience intelligence in real time. Existing literature demonstrates individual components—IoT smart rooms, voice assistants, service robots, recommendation models, acceptance models, and centralized hotel systems—but there is insufficient empirical evidence for a single interoperable architecture that continuously senses hotel state, predicts demand or service needs, personalizes guest interactions, executes routine actions, and intelligently escalates complex cases to staff.

A second gap concerns evaluation. Current work often measures either guest intention/satisfaction or a narrow technical outcome. A stronger research design should jointly measure AI accuracy, response latency, task-completion rate, recommendation quality, resource utilization, energy/service efficiency, guest satisfaction, complaint resolution time, staff workload, privacy/security incidents, and cost. This would enable a meaningful comparison between conventional hotel operations and AI-assisted operations.

7. Proposed Research Direction

- Develop a modular architecture with an integration/API layer connecting PMS, CRM, housekeeping, access control, IoT sensors, smart-room devices, restaurant/service modules, and guest applications.
- Use an event-driven data layer to create a real-time hotel digital state: occupancy, room status, service requests, maintenance alerts, guest preferences, and staff availability.
- Implement predictive models for occupancy, demand, maintenance, housekeeping workload, and service-request prioritization.
- Use an NLP/LLM-based hotel assistant with retrieval from an approved hotel knowledge base, multilingual support, conversation memory limited to necessary context, and explicit escalation to staff.
- Build a recommendation engine combining historical preferences, review sentiment, current context, hotel availability, and guest constraints.
- Introduce human-in-the-loop orchestration: routine low-risk tasks can be automated; ambiguous, sensitive, or high-impact decisions are routed to staff.
- Apply privacy-by-design: data minimization, role-based access, encryption, retention policies, consent management, and non-identifying analytics wherever possible.
- Evaluate the prototype through a controlled pilot or simulation with a common KPI framework and compare it with baseline hotel workflows.

8. Suggested Evaluation Framework

Dimension	Metric	Baseline	Target/interpretation
AI/NLP performance	Intent accuracy, F1,	Rule-based/FAQ baseline	Higher accuracy and

	answer groundedness, fallback rate		groundedness; fewer unsafe/incorrect answers
Responsiveness	Median and P95 response latency	Current manual/legacy workflow	Lower latency without reducing answer quality
Operations	Task completion rate, service-request resolution time	Manual hotel process	Faster resolution and fewer missed requests
Prediction	MAE/RMSE/MAPE for demand/occupancy forecasts	Naive/seasonal baseline	Lower forecast error
Recommendation	Precision@K, Recall@K, NDCG, acceptance/click/use rate	Popularity or basic collaborative baseline	Better personalized ranking
Guest experience	CSAT, NPS, perceived usefulness, effort, complaint rate	Conventional service	Improved satisfaction and reduced effort
Staff impact	Requests per staff hour, escalation rate, workload	Manual workflow	Lower routine workload while preserving service quality
Efficiency	Energy/room utilization, operational cost per occupied room	Existing operations	Lower resource consumption/cost
Security/privacy	Access violations, privacy incidents, data-retention compliance	Existing controls	No critical incidents; measurable compliance
Scalability	Concurrent users/devices, throughput, failure recovery	Single-site prototype	Stable performance as hotels/devices/users increase

9. Conclusion

The reviewed literature establishes that AI-based smart hospitality is technically feasible and increasingly relevant, but the evidence base is still divided among guest acceptance, service robotics, voice interaction, recommendation, and isolated operational systems. The most promising direction is an integrated architecture that treats the hotel as a real-time intelligent service ecosystem. Such a system should combine IoT and enterprise data with predictive analytics, personalized recommendation, conversational AI, and human-in-the-loop orchestration. The proposed research can make a stronger contribution if it demonstrates the architecture in a realistic pilot or testbed and evaluates both technical performance and actual guest/employee outcomes using standardized metrics.